Pan-Tilt Camera Mechanism Report

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Objective

The objective of this project is to design and develop a compact, lightweight, and fully 3D-printable pan-tilt camera mechanism for an ESP32-CAM using two SG90 servo motors. The mechanism is designed to provide approximately 180° pan and 90° tilt motion, enabling the camera to move in both horizontal and vertical directions. The design focuses on simplicity, ease of manufacturing using FDM 3D printing, smooth operation, and suitability for robotics, surveillance, and vision-based applications.

Working Principle

The pan-tilt camera mechanism operates using two SG90 servo motors that provide motion along two independent axes. The lower servo controls the pan movement by rotating the entire upper assembly horizontally through approximately 180°. The upper servo controls the tilt movement by rotating the camera mount vertically through approximately 90°. The ESP32-CAM is mounted on the camera plate attached to the tilt mechanism, allowing the camera to be positioned in different directions for a wider field of view. The mechanism enables smooth and controlled camera orientation, making it suitable for robotics, surveillance, object tracking, and other vision-based applications. All structural components are designed to be manufactured using FDM 3D printing, resulting in a lightweight and compact assembly.

Design Features

- Two-axis pan-tilt mechanism
- Designed for ESP32-CAM
- Compatible with SG90/MG90S servos
- Fully 3D-printable parts
- Compact and lightweight design
- Simple assembly

Printing Orientation

The components are oriented to ensure easy manufacturing using an FDM 3D printer while minimizing the need for support material. The base plate is printed flat on the build plate to provide maximum stability and dimensional accuracy. The camera mounting plate is also printed flat to achieve a smooth surface finish and accurate mounting features. The servo bracket and other supporting components are oriented to reduce overhangs and improve structural strength. This printing orientation helps minimize printing time, material usage, and post-processing while ensuring good mechanical performance of the assembled mechanism.

Tolerances

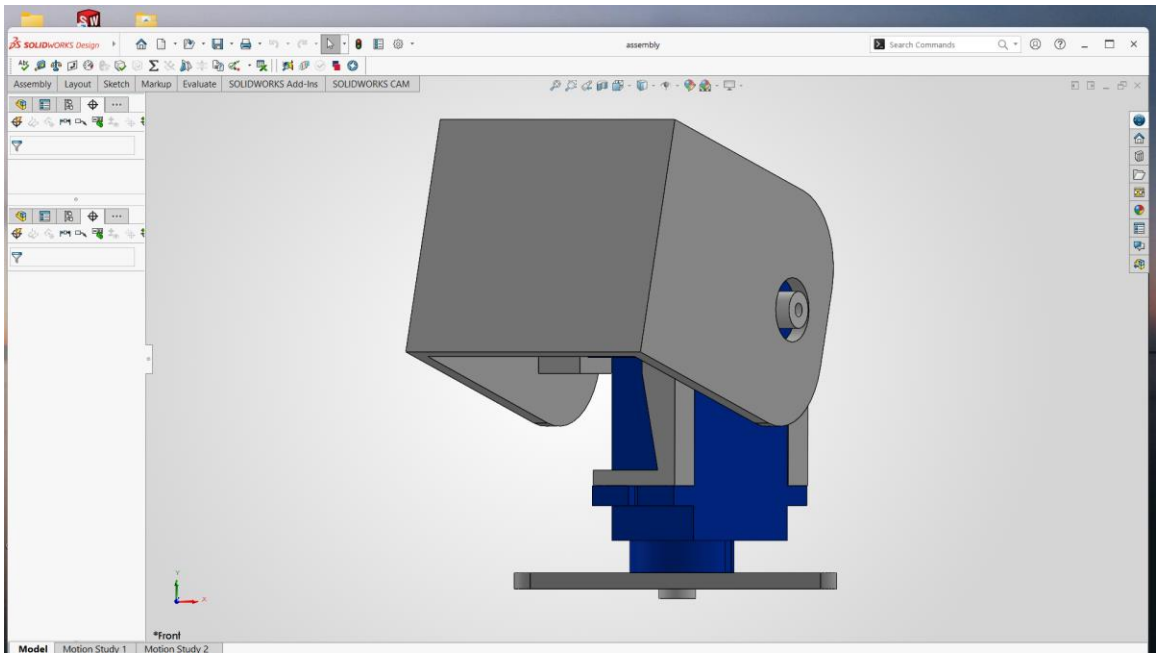
A dimensional clearance of 0.2–0.3 mm is maintained between mating 3D-printed parts to ensure smooth assembly and movement after FDM printing. This tolerance compensates for printing inaccuracies, material shrinkage, and surface finish variations, allowing the moving components to operate without excessive friction while maintaining adequate structural stability.

Assumptions

- Standard dimensions of the SG90 servo motor and ESP32-CAM module are considered during the design process.
- All structural components are assumed to be manufactured using PLA material through FDM 3D printing.
- A printing clearance of 0.2–0.3 mm is assumed to ensure proper fit and smooth movement between mating parts.
- The mechanism is intended for lightweight camera applications and is not designed to carry heavy loads.
- Servo motors are assumed to provide sufficient torque for smooth pan and tilt movement under normal operating conditions..

Bill of Materials

Component	Quantity
Base Plate	1
Servo Bracket	1
Camera Plate	1
Pan-Tilt Link	1
SG90 Servo Motor	2
ESP32-CAM Module	1



Conclusion

The designed pan-tilt camera mechanism successfully provides two-axis motion for the ESP32-CAM using two SG90 servo motors. The model is compact, lightweight, and suitable for FDM 3D printing. The design demonstrates the principles of CAD modeling, assembly, and motion required for camera positioning applications. This mechanism can be used in robotics, surveillance, object tracking, and other vision-based systems, and can be further enhanced with control electronics and automation for real-time operation..